IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

SELECTED TOPICS IN POWER SYSTEMS

Wednesday, 7 May 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Answer ALL Questions

Question 1 on Flexible AC Transmission System (FACTS)

a) Consider two areas interconnected by two different transmission routes. The reactance of one route is twice that of the other. In case of frequent congestion (or overloading) of the route with lower reactance, explain why upgrading this route by adding another similar line in parallel is not necessarily the best option.

[2]

b) A 275 kV transmission line carries 3.0 p.u. active power. The line reactance is 0.25 p.u. The voltages at both ends are maintained at 1.0 p.u. Considering a three-phase short circuit fault on the line that clears automatically, determine the critical clearing angle (in degrees) for the base case without any FACTS device installed on the line. Neglect the resistance and shunt susceptance of the line and consider zero power transfer during the fault.

[2]

For the following sub-questions consider the same transmission line and active power flow as in part (b) and only one FACTS device at a time. Use the same assumptions.

c) A Thyristor-Controlled Series Capacitor (TCSC) is installed on the transmission line to provide 20% series compensation.

i) Determine the new critical clearing angle (in degrees).

[3]

ii) How much active power can be transmitted with 20% series compensation if the critical clearing angle were to be the same as in the base case in part (b)?

[3]

d) A Thyristor-Controlled Phase Angle Regulator (TCPAR) is installed on a transmission line to enhance transient stability with an ideal phase angle compensation in the range of ±18°. For an initial phase angle setting of 4°, determine the steady-state stability limit.

[2]

e) A Static VAr Compensator (SVC) with a rated susceptance of 5.0 p.u. is installed at the midpoint of the transmission line to maintain the voltage at 1.0 p.u. Determine the critical clearing angle (in degrees).

Reactive power exchanged with the SVC

$$Q_C = \frac{4}{X}\left(1 - \cos\frac{\delta}{2}\right)$$

Power-angle relationship once the rated susceptance limit (B_{max}) of the SVC is reached

$$P = \frac{4}{X(4 - XB_{max})}\sin\delta$$

[8]

[20 marks]

Question 2 on System Operation with Low Inertia

a) To maintain frequency stability in the future low-inertia power system, it is important to meet the system frequency requirements after large disturbances. The system dynamic is complicated but can be approximately described by the following swing equation:

$$2H\dot{f}(t) = R(t) - \Delta P_L$$

i) Describe the three frequency constraints that need to be maintained during system operation, and identify the key factors that affect those constraints.

[3]

ii) Analytically derive the frequency nadir (at time instant t_n) constraint in terms of H (system inertia in [s]), R (frequency response linearly delivered by T_d) and Δf_{max} from the swing equation above and assume $t_n < T_d$.

[4]

iii) The constraint that guarantees the system frequency nadir within a maximum admissible frequency deviation is given below:

$$\left(H - \mathrm{DC}\times\frac{\mathrm{T_{DC}}}{4\times\Delta f_{\max}}\right)\times R \geq \frac{(P_{\mathrm{L}} - \mathrm{DC})^2 \times \mathrm{T_d}}{4\times\Delta f_{\max}}$$

Nomenclature

H:	System level of inertia (MWs/Hz)
R:	Amount of frequency response scheduled (MW)
DC:	Amount of Dynamic Containment (also called Enhance Frequency Response (EFR)) scheduled (MW)
P_{L}:	Value of generation outage we want to secure against (MW)
$\Delta f_{\max}$:	Maximum admissible frequency deviation to avoid load shedding
$\mathrm{T_d}$:	Delivery time of frequency response (s)
$\mathrm{T_{DC}}$:	Delivery time of DC (s)

Other assumptions:

- The largest generation unit is 1.8 GW
- Numerical values: H = 3 GWs/Hz, $\Delta f_{\max}$ = 0.8 Hz, $\mathrm{T_d}$ = 11.2 s, $\mathrm{T_{DC}}$ = 0.64 s

The system operator has procured the **optimal** amount of DC and R to maintain the nadir constraint. In this scenario, it happens that $\boldsymbol{R = 3.5*DC}$. Calculate DC and R that have been procured and the relationship of their marginal prices.

[3]

b) Complete the following tasks related to the implementation of frequency support services in renewable generation systems:

i) A transmission system operator requires renewable generators to implement a synthetic inertia controller that adjusts their power output, ΔP, as a function of the RoCoF. Explain why it depends on the RoCoF and how the emulated inertia time, H, affects the magnitude of ΔP.

[4]

ii) A wind turbine with a radius of 51 m has a rated power of 3 MW. The moment of inertia of the turbine is $I_g = 1.6 \cdot 10^7$ kg m^2. Complete the following tasks:

a. Calculate how much kinetic energy is stored in the turbine when it rotates at a speeds of $\omega_1 =$ 20 min^{-1} and $\omega_2 =$ 18 min^{-1}.

[2]

b. Calculate how much additional power the wind turbine could deliver to the grid for a period of 30 seconds (a constant ΔP) if it slowed down from ω_1 to ω_2? Consider that the power extracted from the wind remains constant during the process.

[2]

iii) Explain why wind turbines aren't designed to maximise their generated power for the range of speeds between rated wind speed and cut-out wind speed.

[2]

[20 marks]

Question 3 on Power System State Estimation

a)

i) What are the states in the context of power system state estimation? [2]

ii) What causes variations in these states? [2]

iii) How does state estimation help to detect stressed system operation? [4]

iv) How does state estimation differ for unbalanced systems compared to balanced systems? [2]

b)

i) What are the typical measurement sets in power distribution state estimation? [2]

ii) How are those measurements classified? [2]

iii) What is a linear state estimator? [3]

iv) What measurement sets can make the linear state estimation possible? [1]

v) What is the most suitable algorithm in power distribution state estimation and why? [2]

[20 marks]

Question 4 on Electricity Market and Peer-to-Peer Trading

a) In an electricity market, there are 4 generation units. The units labelled "S1" and "S2" are owned by two strategic producers, respectively, while "O1" and "O2" belong to non-strategic producers who act as price-takers. The market experiences two demand levels: "D1" during the off-peak period and "D2" during the peak period. The wind farm "O1" operates at 60% of its capacity during the off-peak period and 80% of its capacity during the peak period.

Table 1: Operational characteristics of generation units

Unit	S1	S2	O1	O2
Technology	Coal	Gas	Wind	Gas
Cost (£/MWh)	14.8	19.1	0	26.7
Capacity (MW)	200	300	300	120

Table 2: Demand levels during off-peak and on-peak periods

Demand period	D1	D2
Quantity (MW)	300	720

i) Perfect competition: assume that all producers offer their true cost, draw market demand-supply curves, calculate generation dispatches, market-clearing prices, and generation profits for two periods.

[6]

ii) Strategic offering: 1) assume that S1 offers strategically at 1.5 times its true cost while S2 continues to offer competitively during the off-peak period; 2) assume that S2 offers strategically at 1.6 times its true cost while S1 continues to offer competitively during the peak period. For both periods, draw market demand-supply curves, calculate generation dispatches, market-clearing prices, and generation profits.

[6]

b) In a local energy market, 2 consumers (deficit) and 2 prosumers (surplus) participate in a double-auction market, where the order book is organised in Table 3. The grid 'buy' price and grid 'sell' price are 0.16 £/kWh and 0.02 £/kWh, respectively.

Table 3: Order book of the double-auction market

Consumer/Prosumer	C1	C2	P1	P2
Bid/Offer price (£/kWh)	0.14	0.08	0.04	0.10
Quantity (kW)	3	4	5	4

i) Calculate all trading activities and update the order book per trading activity.

[4]

ii) Calculate the payment of the 2 consumers and the revenue of the 2 prosumers in this local energy market.

[4]

[20 marks]